
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

Anonymous authors
Anonymous submission

ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead on long videos because they must process massive sequences of visual tokens. Existing efficiency methods reduce this burden mainly by pruning or merging tokens according to importance or similarity. This redundancy-centric view leaves a decisive aspect of video understanding under-modeled: changes, turning points, and the spatio-temporal relationships that make those changes meaningful. We introduce ST-SimDiff, a training-free framework built on a clear principle: similarity identifies redundancy, whereas difference identifies key events. ST-SimDiff first constructs a spatio-temporal graph over visual tokens to model their spatial and temporal associations in a unified structure. It then performs parallel dual selection: 1) similarity-based selection applies community detection to retain representative tokens and compress stable content; 2) temporal difference-based selection pinpoints content-changing moments and preserves tokens that encode dynamic shifts. The resulting representation keeps both persistent scene information and event-defining transitions under a compact token budget. Extensive experiments demonstrate that ST-SimDiff decisively outperforms state-of-the-art compression approaches while substantially lowering computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rise of Large Language Models (LLMs) has accelerated the development of Large Vision-Language Models (LVLMs), which now deliver strong image and video understanding capabilities (Zhang et al., 2024c; Liu et al., 2024c). For video processing, current LVLMs typically sample a video into frames and convert each frame into hundreds or thousands of visual tokens for the LLM (Bai et al., 2025; Li et al., 2024). This pipeline is effective, but its token count grows rapidly with video duration and resolution. The resulting computational and storage costs become prohibitive, making efficient token handling a central requirement for long-video analysis and real-time interaction (Fu et al., 2024b).

Recent work improves LVLM efficiency through two dominant routes. Importance-based pruning removes tokens that receive lower attention in later model layers, building on the observation that LVLM attention contains substantial redundancy during visual processing (Chen et al., 2024). Similarity-based merging or selection instead exploits the fact that many visual tokens are highly similar within a frame or across adjacent frames. FrameFusion (Fu et al., 2024b) merges similar tokens between neighboring frames, and VisionZip (Yang et al., 2025) selects dominant tokens directly at the vision-encoder level. Together, these methods establish that visual token redundancy is real and compressible.

Despite this progress, existing token reduction methods leave two important gaps unresolved. First, many approaches treat spatial correlations within a frame and temporal correlations across frames separately, rather than modeling their joint spatio-temporal structure. This separation weakens their ability to preserve complex dynamic events. Second, the field has largely optimized for commonality—importance and similarity—while underusing difference. Video narratives are of-

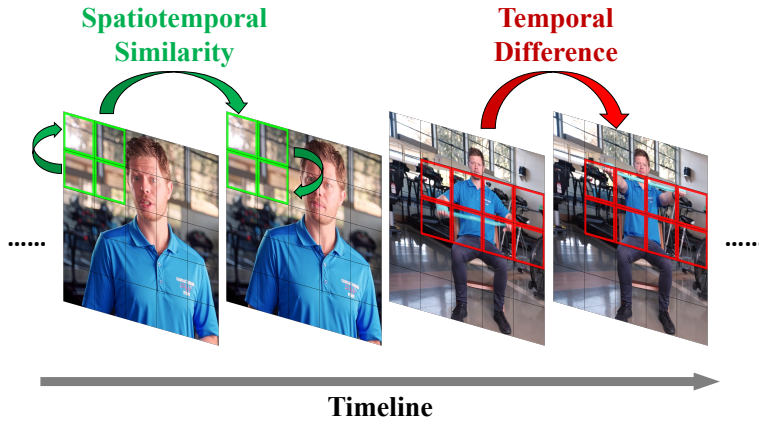


Figure 1: Core motivation of ST-SimDiff. Efficient video understanding must handle two complementary signals: Spatiotemporal Similarity (left) compresses redundant information within frames and across adjacent frames, while Temporal Difference (right) identifies the key actions or events that define the plot.

ten defined by turning events; a compression method that only searches for similarity can smooth over sudden actions and lose the evidence needed for correct interpretation.

We therefore put forward a sharper principle for video token compression: similarity identifies redundancy, while difference captures key moments. An effective compressor must simultaneously summarize stable content with as few tokens as possible and preserve the changes that drive the video. To realize this principle, we propose **ST-SimDiff**, a dual token selection framework built on a spatio-temporal graph. The framework has two coordinated parts. **Similarity-based representative token selection** treats all video tokens as graph nodes and connects them according to feature similarity in the spatio-temporal domain. Community detection then reveals tightly connected token clusters corresponding to stable and persistent visual elements, from which high-centrality tokens are retained as representatives. **Difference-based event token selection** focuses on temporal graph edges. When the similarity between corresponding tokens in adjacent frames drops sharply, ST-SimDiff identifies the transition as a turning point and retains the associated tokens as critical event evidence. The final token set merges these representative and event tokens before input to the LLM. Our contributions can be summarized as follows:

- We establish a similarity–difference perspective for VLM efficiency, arguing that token compression must value redundancy removal and event preservation equally.
- We design a spatio-temporal graph framework that uniformly models complex relationships among video tokens and supports a dual selection strategy for both representative tokens and pivotal event tokens.
- We validate ST-SimDiff through extensive video understanding experiments, showing that it substantially reduces the visual token budget while maintaining or improving performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has firmly established Large Vision Language Models (LVLMs) as a leading architecture for multimodal understanding (Liu et al., 2024b; Bai et al., 2023). These models convert images or video frames into visual tokens with pre-trained visual encoders and feed the resulting token sequence into LLMs. Alignment modules such as MLPs (Liu et al., 2024a;b) and Q-Formers (Li et al., 2023; Dai et al., 2023) connect the visual encoder to the language model. This

design has produced strong multimodal capabilities across tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a), and in video understanding it has enabled models to process increasingly long, high-resolution, and complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

This capability comes with a direct computational cost. Video inputs require continuous frame sequences, so the number of visual tokens far exceeds that of static images and places heavy pressure on both training and inference. Efficiently processing massive video token streams is therefore not a peripheral optimization but a core problem for practical LVLM deployment.

2.2 VISUAL TOKEN COMPRESSION

Because video processing is dominated by token overhead, visual token compression has become a central route to efficient LVLM inference. Existing methods primarily compress from two perspectives: token importance and token similarity.

Early approaches focused on importance-based pruning: they identify visual tokens that contribute less to model performance and remove them. FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), for example, use attention scores or related feature metrics to estimate token importance. Later work (Shang et al., 2024; Ye et al., 2025) recognizes that important tokens can still be redundant when they encode highly similar information. FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) selects dominant tokens while merging contextual information, and VisPruner (Zhang et al., 2025b) uses visual cues to remove redundant items. These methods make token compression effective, but they remain largely organized around preserving common or salient content.

Despite clear progress on visual token redundancy, existing work still underuses the global spatio-temporal similarity structure of video tokens. Video redundancy is both spatial and temporal, and these two dimensions interact. Similarity-driven methods such as FrameFusion mainly emphasize temporal similarity between adjacent frames and do not fully exploit the richer spatio-temporal relationships among tokens. This limitation constrains redundancy removal, especially under aggressive token budgets where every retained token must carry distinct value.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) encodes the video into a full sequence of N visual tokens, $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This sequence is expensive because self-attention scales quadratically with the token count N , i.e., $O(N^2)$. We therefore formulate video token compression as the design of an efficient selection function, $f(\cdot)$, that maps the full sequence T_{full} and a compression ratio r to a compact subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The objective is to maximize downstream LVLM performance while keeping $|T_{\text{sub}}|$ approximately equal to $r \cdot N$. The central design problem is decisive token retention: $f(\cdot)$ must keep the information that matters most for video understanding while discarding redundant visual evidence.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff operates on a spatio-temporal graph built from video visual tokens and uses two parallel components: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). SRTS targets stable and redundant content. It applies community detection to the graph, groups highly similar tokens into clusters that correspond to persistent visual elements such as static backgrounds, and retains high-centrality tokens from each cluster as the representative set T_{rep} . DETS targets dynamic shifts. It examines temporal graph edges and identifies moments where corresponding tokens in adjacent frames sharply lose similarity. These high-difference points mark turning events, so the tokens around them are retained as critical event tokens, forming T_{trans} . By combining these two components, ST-SimDiff produces $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$: a compact token subset that preserves stable scene content without sacrificing event-defining changes.

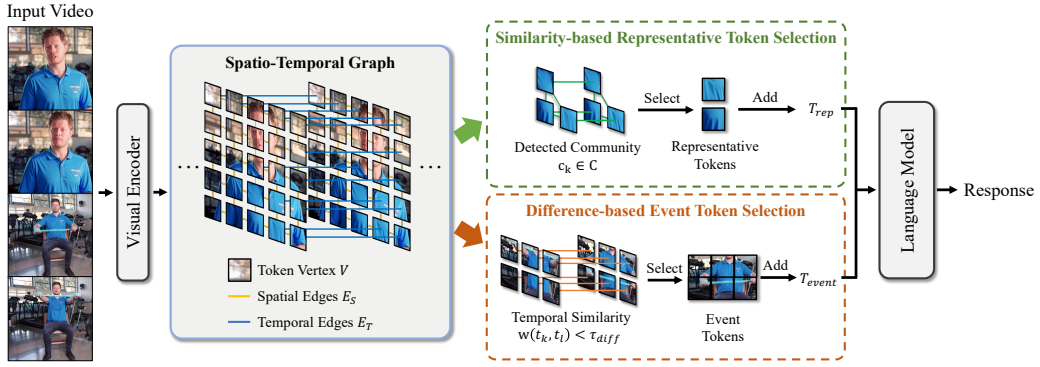


Figure 2: Overview of ST-SimDiff. The framework performs Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection to retain compact yet event-aware video evidence.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

For a given video, the vision encoder produces a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ occupies a specific spatio-temporal location and carries a feature vector $\mathbf{x}_k$. We specify the token location through its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$.

We represent token relationships with a spatio-temporal graph $G = (V, E)$, where the vertex set V contains all tokens in T . The edge set E decomposes into $E = E_S \cup E_T$, explicitly separating spatial adjacency from temporal continuity. The spatial edges E_S connect tokens that are adjacent within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens at the same spatial position in adjacent frames, directly encoding temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

Each edge weight $w(v_i, v_j) \in E$ is the cosine similarity between the corresponding token features:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph efficiently captures local spatial structure and temporal continuity in one representation, providing the foundation that enables our dual-path selection strategy.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content contains substantial spatio-temporal redundancy. Static backgrounds remain highly similar across time, and moving objects can preserve semantic consistency even as their spatial positions change. Existing methods often treat temporal and spatial similarities separately; ST-SimDiff instead captures their joint structure through the spatio-temporal graph. In this graph, semantically related tokens form dense “communities” or “clusters” according to feature similarity, regardless of their exact frame or location. Selecting representatives from these communities is therefore a direct and effective way to remove redundancy while preserving core semantics.

To identify strongly correlated token clusters, we first threshold the graph G . We set a similarity threshold τ_{sim} and retain only edges whose weights exceed this threshold, yielding $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to G' and obtain token clusters $C =$

$\{c_1, c_2, \dots, c_m\}$. Within each community $c_k \in C$, we rank tokens by centrality. The centrality score $S_c(t_a)$ of token $t_a \in c_k$ is its average similarity to all other tokens in that community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

We set the intra-community retention rate to the global compression ratio r and keep the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores in each community c_k . This rule handles all communities uniformly, including single-node communities of unique tokens, where the sole node is retained. Formally, the representative token set T_{rep} is:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score}=S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where TopK returns the k highest-scoring elements from a given set. This strategy compresses static and persistent video content by retaining the most central evidence within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity defines the “norm” of a video, difference defines its “events”. A similarity-only compressor can preserve “what is” while losing “what happened”. Video plots are driven by changes: a new object appears, an action begins, or a scene transitions. In ST-SimDiff, these changes appear as sharp feature differences between temporally adjacent tokens. Capturing such turning points is therefore essential for preserving dynamic content and answering questions such as “when” and “why”.

Temporal difference is a strong signal for key events. The entry of a new object, a scene change, or the start or end of an action produces large feature changes at corresponding positions in adjacent frames. We therefore analyze the temporal edges (E_T) in the spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ connecting adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score exceeds this threshold, i.e., $w(t_k, t_l) < \tau_{\text{diff}}$, we designate the subsequent token t_l as a critical event token and retain it. Formally, the event token set T_{event} is:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This strategy explicitly preserves tokens that mark significant temporal change. By retaining these transitions, ST-SimDiff prevents the compression process from discarding the dynamic evidence that defines the video’s narrative.

4.5 OVERALL REDUCTION PROCESS

ST-SimDiff computes the representative token set T_{rep} and event token set T_{event} in parallel, then forms the candidate set $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$. To exactly meet the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$, we add a final pruning step. If $|T_{\text{candidate}}|$ exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ lowest-importance tokens. Following existing work (Chen et al., 2024), token importance is determined by the attention score in shallow LLM layers. This step lets ST-SimDiff satisfy a prescribed computational budget while retaining the most valuable candidate tokens.

Overall, the method combines graph-based token selection with attention-based dynamic pruning. The graph component uses intrinsic video structure—similarity and difference—to retain core content, while the pruning component enforces the target budget. This design makes token compression principled, adaptive, and strongly aligned with the efficiency–performance trade-off.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff decomposes into three stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video understanding benchmarks using LLaVA-Video-7B with 64 input frames (%). The strongest result among compressed methods is emphasized in **bold**.

Graph construction computes cosine similarity between each of the N visual tokens and its spatially and temporally adjacent neighbors. Because each token has only a small constant number of neighbors, this stage requires a single pass over the token sequence and has complexity $O(Nd)$, where d is the feature dimension.

After graph construction, SRTS identifies token clusters. As detailed in our implementation (Section 5.2), we use a connected components algorithm with near-linear complexity $O(N + |E|)$. Since $|E|$ is at most $3N$, this reduces to $O(N)$. The subsequent within-community filtering could become expensive under a naive all-pairs similarity computation, which would cost $O(|C| \cdot |c_k|^2 \cdot d)$ for community c_k . We avoid this cost by bounding community size; if a community exceeds a pre-defined threshold (e.g., $|c_k| > \sqrt{N}$), we partition it. This keeps filtering efficient and prevents it from dominating the total cost. Although graph clustering methods such as Louvain and Leiden provide richer community definitions, their complexity is higher, approaching $O(N \log N)$. We therefore use connected components to keep the compression overhead minimal.

DETS scans the temporal edges once and identifies significant changes by comparing token similarities against the threshold. This stage is also linear, with complexity $O(Nd)$.

Thus, ST-SimDiff is governed by linear-time operations and has overall complexity $O(Nd)$. This is substantially lower than the quadratic self-attention cost, $O(N^2d)$, in the Large Language Model. The framework therefore delivers significant token reduction with only a small computational footprint relative to the main inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines We evaluate ST-SimDiff on LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c), two strong base models that test whether the framework transfers across architectures. We

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video understanding benchmarks using NVILA-Video-8B with 64 input frames (%). The strongest result among compressed methods is emphasized in **bold**.

compare against state-of-the-art video token compression methods spanning multiple design families: **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), and **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). For a complete reference, we also report the original uncompressed model (Vanilla) as the performance upper bound and Random Sampling with the same token count as a lower bound.

Benchmarks We test long-video understanding on three demanding benchmarks: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME provides broad foundational video understanding coverage with diverse video types and Short, Medium, and Long length categories. LongVideoBench evaluates detailed retrieval and reasoning in long videos through a “referring reasoning” task over 15s, 60s, 600s, and 3600s videos. EgoSchema focuses on long-term first-person question answering and probes whether a model can recover character intentions and causal action chains.

5.2 IMPLEMENTATION DETAILS

All experiments run on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04. We conduct inference evaluation with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow the original 64-frame input setting; for NVILA, we also use 64 frames to keep computational efficiency and resource comparisons aligned. We set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$. We evaluate token retain ratios of $r = 30\%$ and $r = 50\%$. In practice, we use connected component finding as the community detection algorithm because it preserves the method’s performance gains while keeping overhead low.

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component ablation results showing the contribution of similarity selection and difference selection (%).

5.3 COMPARISON WITH STATE-OF-THE-ARTS

We compare ST-SimDiff with mainstream efficient video language model (VLM) compression methods on VideoMME, LongVideoBench, and EgoSchema. To test robustness rather than a single-model fit, we report results on both LLaVA-Video-7B and NVILA-8B under 30% and 50% token retain ratios. Table 1 and Table 2 present the full comparison.

(1) ST-SimDiff consistently delivers the strongest overall results across the tested configurations. At a 50% token retain ratio, our method surpasses competing compression algorithms on both base models and, on several benchmark scores, matches or exceeds the original model that uses 100% of the tokens. These results provide direct empirical support for the ST-SimDiff design: retaining both representative content and event transitions is more effective than redundancy-focused compression alone.

(2) Among importance-based baselines such as FastV and FasterVLM, FasterVLM is typically stronger, but this family still handles video temporal redundancy inefficiently because it can retain important yet repetitive tokens. Hybrid baselines such as PruMerge, VisionZip, and FrameFusion are more competitive, with FrameFusion generally providing the strongest baseline results. Even so, their optimization target remains centered on representative “commonality” information. This design leaves them vulnerable to missing narrative evidence carried by “turning points” and “abrupt events”.

(3) Overall, existing baselines show that the community has increasingly recognized video “redundancy,” but they leave two core challenges unresolved: they do not fully analyze joint spatio-temporal similarity among visual tokens, and they do not actively preserve “difference” signals that define plot development. ST-SimDiff gains its advantage by addressing both issues directly: the spatio-temporal graph models complex token relationships, and difference detection explicitly protects key changes.

5.4 ABLATION STUDY

We isolate the contribution of each ST-SimDiff component through detailed ablations. Starting from a Baseline strategy with only fundamental importance-based pruning, we progressively add the similarity selection module (+Sim) and the difference selection module (++Diff). All ablations use token retain ratios of 30% and 50%, and Table 5 reports the results.

(1) The +Sim module consistently improves the baseline, confirming that similarity modeling is a reliable mechanism for redundancy compression. We further decompose similarity into spatial-only

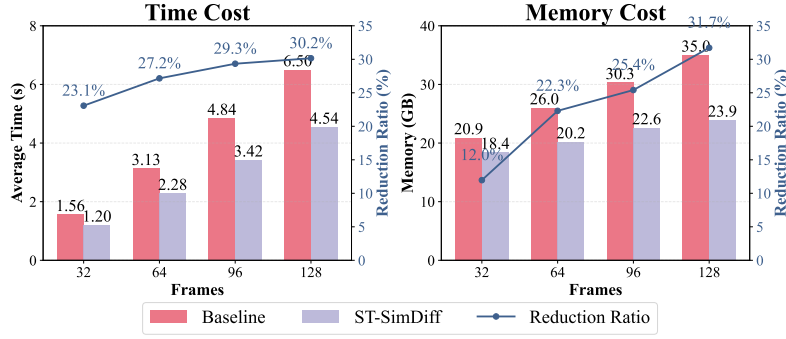


Figure 3: Computational cost comparison between ST-SimDiff and the baseline LLaVA-Video model in average inference time and peak GPU memory usage under a 30% token budget.

(Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) variants. The joint variant performs best, demonstrating that redundancy is most effectively compressed when spatial and temporal associations are modeled together.

(2) Adding ++Diff on top of the best joint similarity module completes the framework and produces a clear gain, especially at the more aggressive 30% retain ratio. This result confirms the complementary role of difference selection: +Sim removes redundancy, while ++Diff identifies and preserves event tokens that define the video’s narrative. The gain is smaller at 50% because the larger token budget allows +Sim to retain more event evidence and overall performance is already close to its upper range. The ablation therefore directly supports our central claim that effective video compression must balance similarity-based redundancy removal with difference-based event preservation.

5.5 COMPUTATIONAL COST ANALYSIS

We quantify the practical efficiency of ST-SimDiff by measuring inference time and peak GPU memory usage under a 30% token budget. The comparison uses the uncompressed Baseline LLaVA-Video model as the reference. Figure 3 reports results for video inputs from 32 to 128 frames.

Inference Time ST-SimDiff provides a clear speedup. As the number of input frames increases, the baseline average inference time rises from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames. ST-SimDiff restrains this growth and processes 128 frames in 4.54 seconds on average. The time-saving rate increases with video length, from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff also substantially reduces memory demand. The baseline peak GPU memory grows from 20.9 GB at 32 frames to 35.0 GB at 128 frames, while ST-SimDiff lowers this pressure by reducing the token sequence passed to the language model. For 128-frame videos, ST-SimDiff uses 23.9 GB peak memory, saving 31.7% relative to the baseline. This reduction makes longer-video processing more feasible on memory-limited hardware.

In summary, ST-SimDiff combines leading benchmark performance with concrete efficiency gains in both time and memory, establishing it as an effective compression framework for video understanding.

6 CONCLUSION

This paper tackles the long-video efficiency bottleneck in Multimodal Large Language Models by addressing both excessive visual token counts and the under-preservation of critical content changes. We propose ST-SimDiff, a training-free framework that builds a spatio-temporal graph to model token associations and applies a dual selection strategy: similarity-based selection identifies redundancy through community detection, while temporal difference-based selection preserves key transition tokens. Experiments on VideoMME and LongVideoBench show that ST-SimDiff reduces inference time and memory usage while consistently outperforming state-of-the-art compression

methods. These results support a clear conclusion: efficient video representation should preserve both representative content and transitional events, and ST-SimDiff provides a practical framework for doing so.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

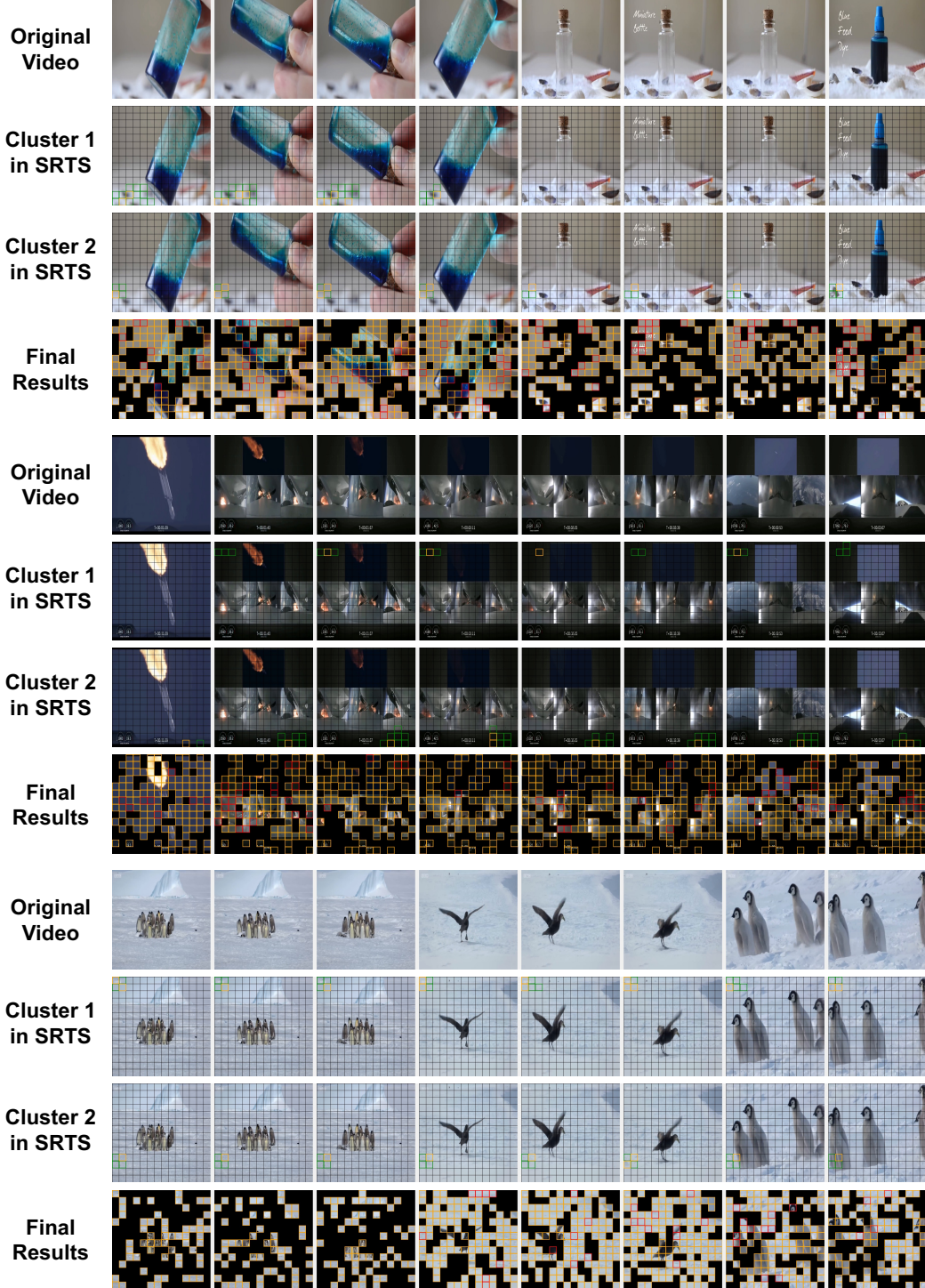


Figure 4: Visualization of the ST-SimDiff token selection process. SRTS identifies token communities (shown as grids in Cluster 1 and 2), while DETS selects event-driven tokens. The final result highlights the intended synergy between sparse representative tokens (yellow) for stable content and dense event tokens (red) for dynamic actions.

Figure 4 provides a concrete visualization of ST-SimDiff’s intermediate and final token selections on a dynamic object manipulation video. The rows labeled “Cluster 1” and “Cluster 2” show the behavior of Similarity-based Representative Token Selection (SRTS). The graph community detection procedure groups spatially and temporally redundant regions, including static background and stationary objects, into coherent communities. The sparse yellow boxes in the final results show that SRTS compresses stable visual elements by retaining only central representative tokens, directly reducing background redundancy.

Difference-based Event Token Selection (DETS) complements SRTS during dynamic moments. In the “Final Results” row, dense red bounding boxes mark tokens triggered by high temporal difference. These red boxes align with the moving hand and shifting blue liquid, illustrating that ST-SimDiff locates “turning points” where temporal difference exceeds the threshold (τ_{diff}). This selection preserves fine-grained action details and rapid visual changes, allowing the model to retain event-defining evidence without being dominated by static content.

The final overlay makes the complementarity explicit: sparse yellow tokens represent stable background evidence, while dense red tokens capture rapid motion. This contrast visually supports the core ST-SimDiff claim. The framework avoids the blind spot of similarity-only compression by allocating tokens to rapid changes, while also pruning spatio-temporal redundancy that importance-based methods can retain. The result is a compact representation that preserves both efficiency and critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video understanding benchmarks using Qwen2.5-VL-7B with 64 and 128 input frames (%).

We further test generalization by applying ST-SimDiff to the recently released Qwen2.5-VL model. These VideoMME and LongVideoBench evaluations examine whether the framework’s gains extend beyond the main base architectures.

Table 4 shows that ST-SimDiff remains effective on Qwen2.5-VL. With a 30% token retention ratio, it reaches 62.4 on VideoMME and 56.8 on LongVideoBench. Increasing the token budget to 50% raises performance to 62.9 on VideoMME and 58.4 on LongVideoBench. At 50%, VideoMME matches the full-token upper bound, and LongVideoBench remains close to full-token performance. These results strengthen the evidence that ST-SimDiff generalizes across base models.

C ABLATION STUDY RESULTS ON NVILA

We also evaluate the generality of each component on NVILA-Video-8B (Liu et al., 2024c). The setup follows the main ablation: we begin with a Baseline strategy using fundamental importance-based pruning, then progressively add the similarity selection module (+ Sim) and the difference selection module (++ Diff).

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	59.3	53.8	50.2
+ Sim	Spatial	59.5	54.5
	Temporal	59.6	54.3
	Spa. + Tem.	59.9	54.6
++ Diff (Proposed)	60.2	55.2	51.7
Token Retain Ratio $r = 50\%$			
Baseline	60.9	55.3	51.5
+ Sim	Spatial	61.1	55.7
	Temporal	61.3	55.6
	Spa. + Tem.	61.4	56.0
++ Diff (Proposed)	61.7	56.5	52.5

Table 5: Component ablation results on the NVILA-Video-8B model (%).

As shown in Table 5, the NVILA ablation reproduces the same pattern observed on LLaVA-Video-7B, reinforcing the framework design. Adding the similarity module (+ Sim) improves performance at both 30% and 50% token retain ratios. Among spatial-only (Spatial), temporal-only (Temporal), and spatio-temporal joint (Spa. + Tem.) variants, joint modeling is the strongest; for example, at $r = 50\%$, it achieves 61.4% on VideoMME and outperforms the separate spatial and temporal variants. This confirms that redundant information is best identified through coordinated spatio-temporal analysis. After adding the best spatio-temporal joint similarity module, introducing the difference selection module (++ Diff) forms the full ST-SimDiff framework and yields the strongest results in the table. This component-level evidence supports the same conclusion across base models: similarity selection and difference selection are both necessary, and their combination is what enables ST-SimDiff to handle redundancy and change together.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

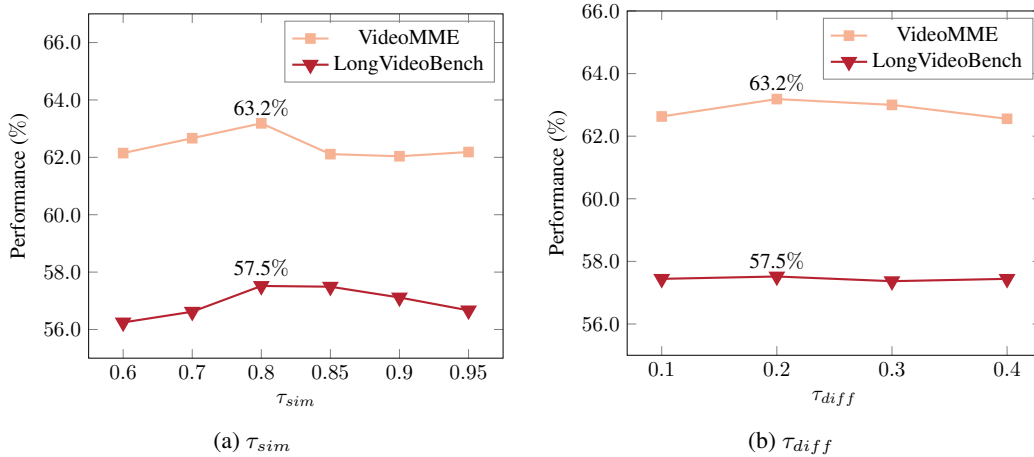


Figure 5: Ablation results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

The similarity threshold τ_{sim} and the difference threshold τ_{diff} directly control representative-token and transition-token selection. Figure 6 shows that both parameters follow a clear rise-then-fall performance trend while remaining robust within a practical range. For τ_{sim} , a threshold that is

too low produces imprecise communities, whereas a threshold that is too high fragments semantic clusters. For τ_{diff} , a threshold that is too low introduces noisy change detections, while a threshold that is too high misses key events. The strongest configuration is $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, so we use this setting in the remaining experiments.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

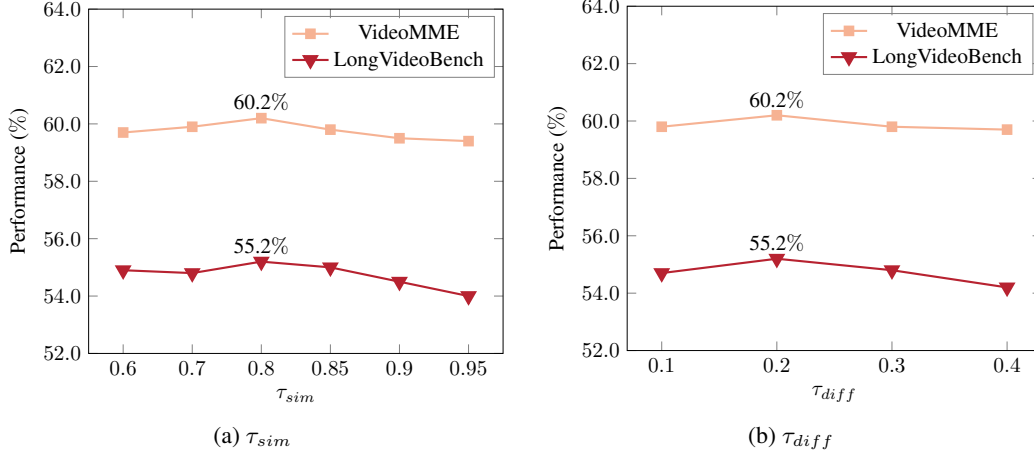


Figure 6: Ablation results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B.

To test hyperparameter robustness on another base model, we conduct ablations on NVILA-Video-8B. As shown in Fig. 6, we evaluate how the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , affect performance on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We test τ_{sim} over $[0.6, 0.95]$. Performance remains stable across this range and peaks at $\tau_{sim} = 0.8$ on both benchmarks, reaching 60.2% on VideoMME and 55.2% on LongVideoBench. Performance declines only slightly when the threshold is too high or too low, showing that ST-SimDiff is not brittle with respect to the similarity threshold.

Impact of Difference Threshold τ_{diff} . We test τ_{diff} over $[0.1, 0.4]$. The model again shows strong robustness, with the best performance around $\tau_{diff} = 0.2$: 60.2% on VideoMME and 55.2% on LongVideoBench. This result indicates that a difference threshold near 0.2 most effectively captures key events while maintaining compression.

In summary, the hyperparameter ablations confirm that ST-SimDiff remains robust to threshold variation and that $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$ are well-supported choices. These values consistently deliver the strongest observed performance across benchmarks and base models.